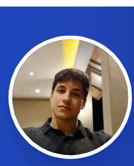
Emir Muhammet Aran



Medical AI Developer | Computer Engineering Student

Medical AI researcher focused on cell imaging and biological aging simulation. Building generative models for label-free microscopy analysis with clinical deployment potential.

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Medium
Ankara, Turkey

Education

Gazi University — Expected 2027
B.S. in Computer Engineering

Work Experience

Developer, ARCx — Applied Research & Community — Present
Solvien

- Medical AI & Computer Vision: Developing MVP projects on Medical Image Processing and generative models
- Large Language Models (LLMs): Implementing state-of-the-art architectures and clinical text analysis
- Open-Source Contribution: Participating in rigorous peer code reviews via GitHub

Software Developer Intern — Summer 2024
Gibrin R&D | TOBB Garaj

- Built real-time IoT monitoring dashboard integrating ESP32 sensors with Firebase cloud database
- Developed secure authentication system with role-based access control across 3 mobile applications
- Implemented cross-platform Flutter applications; shipped to internal users at TOBB Garaj

Medical AI Projects

Spatial Architecture of Pancreas Aging & Senotypes SOLVIEN

Advanced multiplexed immunofluorescence (PhenoCycler/CODEX 41-channel) analysis of the human pancreas to decode the cellular architecture of aging. Unveiled cell-type specific "Bystander Senescence" mechanisms where Alpha cells act as toxic SASP-secreting hubs. Discovered the "Immune Exclusion" barricade and HMGB1 nuclear leakage driving macrophage chemotaxis, actively rewriting the narrative of age-related tissue entropy.

Spatial Analysis | PhenoCycler (CODEX) | Cellular Senescence | Immune Evasion | ONGOING

Article

MSC Bidirectional Aging Simulation — LDM vs CycleGAN

First-author generative pipeline performing label-free, non-destructive morphological "time-travel" on Mesenchymal Stem Cells — translating young↔senescent phenotypes directly from bright-field microscopy (no SA-β-gal staining). Comparative study of Latent Diffusion Models (SDEdit img2img + class conditioning + latent L1 fine-tuning) vs CycleGAN, built on an 8,312 single-cell dataset via Mask R-CNN pseudo-labeling. Champion LDM **outperformed CycleGAN** in both fidelity and training stability: Aging FID **47.30**, Rejuvenation FID **53.49** (5-seed validated, ±0.89/±0.49). Includes ResNet-18 morphology evaluator with Grad-CAM interpretability and an interactive Gradio prototype.

PyTorch | Latent Diffusion | CycleGAN | SDEdit | Mask R-CNN | Bidirectional | Grad-CAM | ✓ FID 47.30

Article | Preprint stage (bioRxiv)

BraTS 2020 Brain Tumor Segmentation

3D medical image segmentation using U-Net with EfficientNetB0 achieving 90.34% Sensitivity and 99.96% Specificity for precise tumor region identification (Necrotic, Enhancing Tumor, Edema). Implements Focal Loss for class imbalance with interactive Gradio interface.

U-Net | 90.34% Sensitivity | 3D Segmentation | Focal Loss | ✓ DEPLOYED

Live Demo | GitHub →

Brain Tumor MRI Classification System

Multi-class deep learning classifier achieving 99% accuracy for brain tumor detection (Glioma, Meningioma, Pituitary, No Tumor). Transfer learning with EfficientNetB3 featuring Grad-CAM visualizations for clinical interpretability.

EfficientNetB3 | 99% Acc | Transfer | Grad-CAM | ✓ DEPLOYED

Live Demo | GitHub →

Breast Cancer Histopathology Detection

Custom lightweight CNN from scratch achieving 91.5% sensitivity (exceeds FDA/EMA screening benchmark) and AUC 0.94. Implemented Focal Loss with optimized clinical threshold prioritizing Recall over Accuracy—catching missed diagnoses. Published detailed analysis on blind augmentation pitfalls and architecture design for medical imaging.

Custom CNN | 91.5% Sensitivity | Focal Loss | FDA/EMA Compliant | ✓ DEPLOYED

Live Demo | GitHub → | Article

Chest X-ray Disease Classification

Multi-label deep learning system for 15-class thoracic disease detection from chest X-ray images using EfficientNetB0 with full fine-tuning. Achieves 0.784 mean AUC with Test-Time Augmentation (TTA). Implements Focal Loss for class imbalance, balanced sampling with oversampling for rare diseases, and medical-appropriate threshold optimization for ≥80% recall priority.

EfficientNetB0 | 0.784 AUC | Multi-Label (15 diseases) | Focal Loss | TTA | ✓ DEPLOYED

Live Demo | GitHub → | Article

EV Smart Parking & Anomaly Detection TÜBİTAK 2209-B

Developed an AI-powered smart parking and security system for electric vehicle charging stations using YOLOv8m. Achieved 96.7% mAP50 for 8-class detection (vehicle, charging status, parking violations, dropped cables). Completely trained on a synthetic dataset generated via NVIDIA Isaac Sim with robust weather augmentation.

YOLOv8 | 96.7% mAP50 | Computer Vision | NVIDIA Isaac Sim | ONGOING

GitHub →

Additional Technical Projects

Clinical NLP & Sequence Architectures
Unstructured clinical text structuring (BioBERT) and automatic subtitle translation (Transformer seq2seq, attention mechanisms)

Computer Vision & Infrastructure
High-throughput DICOM image processing pipelines and real-time face recognition (OpenCV + MTCNN pipeline)

Predictive Analytics
Ensemble methods (Random Forest, XGBoost) and time series forecasting (FBProphet) for optimization tasks

View all projects on GitHub (54 repositories)

Technical Skills

Programming Languages
Python, C, C#, Java, SQL, HTML/CSS, Bash, Arduino

AI/ML Frameworks
PyTorch, MONAI, TensorFlow/Keras, Scikit-learn, OpenCV, XGBoost, HuggingFace Transformers, NVIDIA Isaac Sim, LLMs

Data Science & Analysis
Pandas, NumPy, Matplotlib, Feature Engineering, Vector Embeddings, Statistical Analysis, LLM Fine-tuning

Development & Deployment
HuggingFace Spaces, REST APIs, Git, Linux, Docker, PostgreSQL, FastAPI, Flask

Achievements & Leadership

- **NeuroBridge AI Hackathon 2026** — Built a multimodal Post-COVID neurological biomarker detection system combining brain MRI radiomics, MoCA cognitive scores, and peripheral blood RNA gene expression.
- **ACUHIT Healthcare Hackathon 2026** — Sole AI/ML developer; architected full-stack clinical NLP system (Sentence Transformer + TF-IDF ensemble) with Flask API and React frontend
- **RARE 26 Grand Challenge** — Developed a Self-Supervised Dual-Branch (GastroNet + DINOv3) vision architecture for predicting neoplastic lesions in endoscopy.
- **Vesuvius Challenge** — Top 39% | 3D volumetric segmentation of ancient carbonized scrolls
- **Ostim Game Jam 2025** - 1st Place Winner | Led team in rapid game prototype development
- **MEDCODES** — Board Member (2024-Present) | Leading technical training sessions for medical engineering students
- **CYBERMEDU** — Board Member (2023-Present) | Organizing cybersecurity and medical technology education initiatives

Languages

English: C1